Insurance Customer Growth Analytics

A Business Analytics Project

Prepared as a Data Analytics Portfolio Project

Industry: Insurance Analytics

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Project Background

Business Context

Insurance organizations generate large volumes of transaction data in their daily operations. These records contain valuable information about customers, insurance products, purchasing behavior, premium-related metrics, and sales activities. By applying data analytics methods, transaction-level insurance data can be transformed into meaningful insights about customer behavior, product performance, and potential growth opportunities.

However, raw transaction data alone does not directly answer important business questions. Organizations need analytical approaches to understand customer value, identify purchasing patterns, evaluate retention behavior, and discover opportunities for improving customer engagement. Since insurance data is often recorded at the transaction or policy level, additional data preparation and customer-level feature construction are required before meaningful analysis can be conducted.

This project explores how business analytics techniques can be applied to insurance transaction data to better understand customer segments, purchasing behaviors, retention patterns, and growth opportunities. Through data cleaning, exploratory analysis, and customer-level analysis, the project aims to demonstrate how data-driven approaches can support strategic decision-making.

Problem Statement

Although a large amount of transactional data has been accumulated, raw data alone does not directly answer key business questions. For example, it is not immediately clear which customers contribute the most value, how customer purchasing behavior differs across product categories, whether repeat customers generate more business value than newly acquired customers, or where cross-selling opportunities may exist.

In addition, the dataset's structure presents several analytical challenges. The data is stored at the transaction level rather than the customer level, which means that customer-level metrics must first be constructed before retention, acquisition, and customer value can be properly evaluated. Some variables are also incomplete or irrelevant to the current business focus and will need to be reviewed and filtered during the data preparation stage.

Project Purpose

The purpose of this project is to transform raw insurance transaction data into actionable business insights through data analytics. Specifically, the analysis focuses on understanding customer value, purchasing behavior, retention patterns, and potential cross-selling opportunities.

By combining data preparation, exploratory data analysis, customer-level feature construction, and visualization, this project aims to identify meaningful customer segments, evaluate differences between customer groups, and develop data-driven recommendations for customer growth and engagement strategies.

Project Objectives & Business Questions

Project Objectives

The primary objective of this project is to use insurance transaction data to generate business insights that can support the customer growth strategy in the insurance industry. More specifically, the project aims to identify customer segments that contribute the greatest business value, understand how customers purchase insurance products, and evaluate whether future growth efforts should focus more on retaining existing customers or acquiring new ones.

From a business analytics perspective, the project also seeks to convert raw transaction-level records into a structured, customer-level analytical dataset for customer value analysis, retention analysis, and cross-selling opportunity identification. The final goal is to provide practical, data-driven recommendations to support future decisions on customer management, marketing strategy, and product growth.

Business Objectives

This project is designed around four core business objectives:

1. Identify high-value customer segments based on premium contribution, purchase frequency, and product ownership patterns.
2. Understand customer purchasing behavior across different insurance products, customer types, and sales-related dimensions.
3. Evaluate the relative business value of new and existing customers in order to inform future growth strategy and resource allocation.
4. Generate actionable recommendations for customer retention, customer acquisition, and cross-selling opportunities.

Business Questions

To achieve the objectives above, the analysis will focus on the following business questions:

Q1. Who are the most valuable customers?

Q2. How do customers purchase insurance products?

Q3. Where are the cross-selling opportunities?

Q4. What factors are associated with customer retention or repeat purchase?

Q5. Should the company prioritize customer acquisition or customer retention?

Dataset Description

Data Summary

This project uses anonymized insurance transaction data to explore customer behavior, purchasing patterns, and growth opportunities through data analytics. The dataset contains transaction-level records with information related to customer characteristics, insurance products, purchase behavior, and business-related attributes.

Each record represents an insurance transaction or policy-related activity. Since transaction-level data does not directly represent complete customer behavior, additional customer-level features are constructed during the analysis process to evaluate customer value, retention patterns, and purchasing trends.

Dataset Overview

Item	Description
Dataset Type	Insurance transaction records
Unit of Observation	Individual policy transactions
Main Focus	Customer value, purchasing behavior, retention, growth opportunities

Key Variables

The variables most relevant to this project can be grouped into four categories:

- Customer-related attributes: General customer characteristics and behavioral indicators used for customer segmentation and analysis.
- Transaction information: Purchase-related metrics used to evaluate customer activity, value contribution, and purchasing patterns.
- Product information: Insurance product categories and product-related attributes used to examine product performance and cross-selling opportunities.
- Retention indicators: Behavioral measures used to analyze repeat purchases, customer retention patterns, and long-term engagement.

Data Notes

Because insurance data is often collected at the transaction level, meaningful customer insights require additional data preparation. The analysis process includes reviewing data quality, removing irrelevant fields, standardizing variables, and constructing customer-level features.

Metrics such as customer value, repeat purchase behavior, and retention patterns cannot always be directly observed from individual transactions. Therefore, feature engineering and aggregation methods are applied to transform raw records into an analytical dataset suitable for exploratory analysis and business insight generation.

Project Plan

Analytical Workflow

This project will be carried out in five main stages. First, the dataset will be reviewed to understand its structure, identify key variables, and assess overall data quality. Second, data cleaning will be performed to remove irrelevant fields, standardize key variables, and prepare a usable analytical dataset. Third, transaction-level records will be aggregated to the customer level in order to support customer value, retention, and repeat purchase analysis. Fourth, exploratory and business-focused analyses will be conducted to evaluate customer behavior, product patterns, and growth opportunities. Finally, the findings will be translated into business recommendations and summarized in a final report and presentation.

Expected Deliverables

The main deliverables of this project include:

- A cleaned transaction-level dataset
- A customer-level analytical dataset
- Python notebooks for data cleaning and analysis
- Summary tables and visualizations
- Business insights and growth recommendations
- A final project presentation

Timeline

Stage	Focus
Stage 1	Business understanding and project scoping
Stage 2	Data review and cleaning
Stage 3	Customer-level dataset construction
Stage 4	Exploratory and business analysis
Stage 5	Recommendations and final presentation

Success Metrics

The project will be considered successful if it can produce both meaningful analytical outputs and actionable business insights. In particular, the project aims to identify clear customer growth opportunities, build a reliable customer-level analytical dataset, and generate practical recommendations for customer retention, acquisition, and product strategy. From a technical perspective, success also includes developing reproducible analysis notebooks and producing a final set of visuals and summary findings that can support future business decision-making.

Preliminary Findings

Initial analysis of the transaction sample suggests that **renewal customers and non-renewal customers play different roles in business growth**. Although non-renewal customers account for more transactions and more unique customers in the sample, renewal customers generate **higher premium value per customer**. Renewal customers contributed nearly half of the total premium despite representing a smaller customer base, indicating that they are a valuable segment for retention and cross-sell efforts.

At the product level, the sample also suggests that **different products support different growth objectives**. Certain product categories demonstrated stronger retention patterns, suggesting opportunities for targeted customer engagement and cross-selling strategies.

These preliminary findings suggest that the insurance industry may benefit from a **dual-track growth strategy**. First, the company can strengthen retention efforts for high-value renewal customers, especially in product lines where renewal premiums are consistently strong. Second, the company can continue to use high-volume non-renewal products as entry points for customer acquisition while exploring opportunities to convert those customers into repeat purchasers over time.

Because the current analysis is based on a transaction sample rather than the full historical dataset, these findings should be treated as **preliminary directional insights rather than final business conclusions**. In particular, the product-level patterns observed in the sample may not fully represent the complete portfolio mix. A larger customer-level dataset and a longer time horizon would be needed to validate whether these patterns hold consistently across products, channels, and customer groups.

Limitations & Next Steps

Limitations

This project provides an exploratory analysis of insurance transaction data and focuses on identifying potential patterns in customer behavior, purchasing activity, and retention trends. Since transaction-level data does not always capture the complete customer lifecycle, additional feature development and longer-term customer history would improve the depth of future analysis.

Some analytical limitations include the need for further validation of certain metrics, refinement of customer-level features, and additional investigation of how different factors contribute to customer engagement and repeat purchasing behavior.

Because the analysis focuses on descriptive and exploratory insights, the findings should be interpreted as preliminary observations rather than definitive conclusions. Additional data and longitudinal analysis would be valuable for validating observed patterns.

Analytical Workflow

Future extensions of this project could include developing a more comprehensive customer analytics framework by:

- Expanding customer-level analysis with additional historical information
- Evaluating long-term customer value and engagement patterns
- Exploring customer segmentation approaches
- Comparing different factors associated with retention and repeat purchasing behavior
- Developing clearer visualization and reporting methods for communicating analytical insights

The long-term goal of this analytical approach is to demonstrate how structured data analysis can transform transaction-level information into meaningful insights and support data-informed decision-making.

Portfolio Note

I was working as a data analysis intern at an insurance company this summer in Shenzhen, China. This project was originally developed using the company's real-world insurance transaction data as part of an analytics project. To respect data confidentiality and privacy requirements, proprietary details and sensitive dataset attributes have been generalized or removed in this public portfolio version. The analytical approach, workflow, and key learning outcomes are preserved to demonstrate the data analysis process.